

Physical Therapy Clinic Management Software

Requirements:

1. Customization including Clinic Logo, Clinic Name, Clinic Address Details, Clinic Contacts, Website, Google Business Details (any other if possible).
2. Admin and User Accounts (Login) Admins can have rights to edit, modify and export data. (Register Doctors, Edit Audit Reports etc.)
3. Users with access to Patient Registration, Patient Appointment Scheduling, Patient Searching, Patient Bill Generation et.
4. Main Panel with Following Tabs
 - a. Clinic Details
 - b. Doctors Tab (Having Dr. Name, Education, Designation, Specialty, Session Charges, Available days and timing)
 - c. Patients Tab (Patient Registration Panel, Patient Search, Patient Bill Details, Patients Visit Details)
 - d. Bill Generation Tab (Ability to select the patient's bill based on the sessions attended) The Bill Must have Clinic Name, Logo and Details of the Clinic, Doctor and the charges, any discount applied etc.)
 - e. Finance Tab (Total Income of the clinic, with filters of date, doctor, month etc.)
 - f. Expenses Tab (Total Expenses with free titles and amounts.
 - g. Dashboard (to display all the information on a single page, can be customizable.
 - h. About Software Tab (with details of the developer and any other as needed.)
5. It should be an offline installer-based software with the ability to export the database details to excel or csv as feasible.

AI Generated Layout

Below is a tightened, implementation-ready Level-1 specification that keeps your scope “basic but complete,” while setting clean seams for Level-2 and Level-3 upgrades.

Level-1 Scope (Essential Offline Clinic App)

1) Core Principles

- **Single-user desktop app with optional additional user logins** (no networking; one local database file).
- **Deterministic workflows:** Register → Schedule → Record visit → Bill → Record payment → View dashboard.
- **Portable data:** One SQLite (or Access) file + one “/exports” folder for CSV/Excel.
- **No background services;** no internet required.

2) Navigation and Screens

Top bar: Dashboard | Patients | Appointments | Visits | Billing | Finance | Expenses | Doctors | Clinic Setup | About

- Use a **left filter panel** + **right data grid** on all list pages.
- Use **modal forms** for add/edit to keep screens uncluttered.
- Provide a **global search** (Ctrl+K) for Patient, MRN, Phone, Invoice, and Appointment ID.

3) Roles and Permissions (Level-1)

Feature	Admin	Front Desk (User)
Clinic Setup (branding, fees)	✓	✗
Doctor CRUD	✓	✗
Patient CRUD	✓	✓
Appointments CRUD	✓	✓
Visits (check-in, notes minimal)	✓	✓
Invoice & Payments	✓	✓
Finance & Expenses	✓	View + Add only
Data Export (CSV/Excel)	✓	✗
Audit Log (view)	✓	✗

Authentication: simple username + password (stored as salted hash). Password reset by Admin only.

4) Data Model (Tables and Key Fields)

Settings

- ClinicName, LogoPath, Address1, Address2, City, Phone, WhatsApp, Email, Website, GoogleBusinessURL
- InvoicePrefix, DefaultCurrency, TaxPercent (nullable), DiscountPolicyText
- BackupPathDefault

Users

- UserID (PK), Username (unique), PasswordHash, Role (Admin|User), ActiveFlag, CreatedAt

Doctors

- DoctorID (PK), FullName, Degrees, Designation, Specialty, SessionCharge, DaysAvailable (JSON), TimeFrom, TimeTo, ActiveFlag

Patients

- PatientID (PK), MRN (unique, auto), FirstName, LastName, Sex, DOB, Phone, WhatsApp, Email, Address, City, EmergencyContact, Notes, CreatedAt

Appointments

- AppointmentID (PK), PatientID (FK), DoctorID (FK), Date, StartTime, EndTime, Status (Booked|Checked-in|No-Show|Cancelled), Reason, CreatedAt

Visits

- VisitID (PK), AppointmentID (FK, nullable), PatientID (FK), DoctorID (FK), VisitDate, SessionType (Eval|Follow-up), SessionCountIndex, NotesBrief, Status (Completed|Cancelled)

Invoices

- InvoiceID (PK), InvoiceNo (unique), PatientID (FK), DoctorID (FK), VisitID (FK, nullable), InvoiceDate, Subtotal, DiscountAmount, TaxAmount, Total, Status (Unpaid|PartiallyPaid|Paid|Void)

InvoiceItems

- ItemID (PK), InvoiceID (FK), ItemCode, ItemDescription, Quantity, UnitPrice, LineTotal
 - Level-1 items: “Physiotherapy Session”, “Initial Evaluation”, “Taping”, “Electrotherapy”, “Home Exercise Program Setup”.

Payments

- PaymentID (PK), InvoiceID (FK), PaymentDate, Method (Cash|Card|Bank), Amount, ReferenceNo, Notes

Expenses

- ExpenseID (PK), ExpenseDate, Title, Category (Rent|Utilities|Supplies|Salary|Other), Amount, PaidTo, Notes

AuditLog

- AuditID (PK), UserID, Action (Create|Update|Delete|Export|Login), Entity (Patient|Appointment|Invoice|...), EntityID, Timestamp

Lookups (optional now, useful later)

- Services (Code, Name, DefaultPrice, ActiveFlag)

All monetary fields use **two-decimal fixed precision**; do not use floating point for currency.

5) Validation Rules (Level-1)

- **Patients:** Phone or WhatsApp required; DOB cannot be in the future.
- **Appointments:** Overlap validator per doctor (no time collision); Date cannot be in the past for new bookings; status transitions constrained.
- **Visits:** VisitDate $\geq$ Appointment.Date (if linked).
- **Invoices:** Total = $\Sigma(\text{LineTotal}) - \text{Discount} + \text{Tax}$; cannot mark **Paid** unless $\Sigma(\text{Payments}) = \text{Total}$.
- **Payments:** Amount > 0 ; cannot exceed outstanding balance.
- **Expenses:** Amount > 0 ; ExpenseDate $\leq$ today.
- **Users:** Unique username; password minimum length policy.

6) Workflows

A. Intake to Billing

1. Register patient $\rightarrow$ auto-assign MRN.
2. Book appointment (doctor, date, time; optional reason).
3. Check-in $\rightarrow$ create Visit.
4. Generate invoice from Visit:
 - Pre-fill **Doctor** and **default SessionCharge**.
 - Allow adding other items and percent/flat discount.
5. Record payment (full or partial).
6. Print/Export invoice (PDF) or share via WhatsApp number saved.

B. Same-Day Walk-in

- Create Patient $\rightarrow$ Visit (no appointment) $\rightarrow$ Invoice $\rightarrow$ Payment.

C. Month-end Finance

- Run Finance tab filters (date range, doctor) → Export CSV for Income and Expenses → Basic net summary on screen.

7) Screen-by-Screen Fields

Clinic Setup

- Branding upload (PNG/JPG), details, invoice prefix, tax %, discount policy text.
- Button: “Test print invoice header”.

Doctors

- Grid with inline edit for SessionCharge, available days (checkboxes Mon–Sun), and time range.

Patients

- Card form; duplicate detector using Phone + Name.
- Actions: “New Appointment”, “New Visit”, “Billing History”, “Export Patient CSV”.

Appointments

- Calendar (day/week) + grid. Quick actions: Check-in, Reschedule, Cancel, No-Show.

Visits

- Minimal form: date, doctor, notes (single paragraph), session type, optional session index.
- Button: “Generate Invoice”.

Billing

- Invoice list with status chips.
- Invoice editor: items table, discount (flat or %), tax toggle, payments pane.
- Print template: logo, clinic details, patient details, itemized charges, discount, tax, total, payment summary, signature line, “Thank you” note.

Finance

- KPIs: Total Invoices, Collected, Outstanding, Average Ticket.
- Filters: date range, doctor, payment method, status.
- Quick reports: “Income by Doctor”, “Unpaid Invoices”, “Collected vs Issued”.

Expenses

- Simple entry form + list. Export button.

Dashboard

- Today's appointments (with status), quick totals (Today's Income, This Month Income, Expenses This Month), outstanding invoices list (top 10), and a basic appointments heat strip by hour.

About

- Developer, version, license, support contacts, "Open logs folder".

8) Exports and Printing (Level-1)

- **CSV/Excel** for Patients, Appointments, Invoices, Payments, Expenses, Finance summaries.
- **PDF** for Invoices only (Level-1).
- File naming: Entity_YYYY-MM-DD_HHMM.csv.
- All exports record an **AuditLog** entry.

9) Offline-first Packaging

- **Database:** SQLite file clinic.db (or Access if you prefer). Keep a single file under /data.
- **Installer:** One offline installer that creates folders:
 - .../ClinicApp/data, .../ClinicApp/exports, .../ClinicApp/logs, .../ClinicApp/backups.
- **Backup:** "Backup now" button → copies clinic.db to /backups/clinic_YYYY-MM-DD.db.

10) Minimal Reports (Level-1)

- **Income by Doctor** (date range): Doctor, Count of Invoices, Subtotal, Discounts, Tax, Total, Collected, Outstanding.
- **Unpaid Invoices:** InvoiceNo, Patient, Doctor, Date, Total, Paid, Balance.
- **Expense Summary:** Category totals in range.

11) Default Values and Automations

- Auto MRN format: CLN-{YYYY}{0000}.
- Auto InvoiceNo: {InvoicePrefix}-{YYYY}-{000000}.
- Default currency symbol from Settings.
- Appointment duration default (e.g., 30 minutes) from Settings.
- Pre-fill SessionCharge from Doctor on new invoice item "Physiotherapy Session".

12) Error Handling and Logging

- User-friendly validation messages at field level.
- A rolling text log in /logs/app.log.

- Graceful failure on locked database files with retry guidance.

13) Security (Right-sized for Level-1)

- Passwords stored as salted hash.
- Local database file access controlled by operating system permissions.
- Admin-only exports and backups.
- Obvious “Show password” toggle on login disabled by default.

14) Ready Hooks for Level-2/3 (no implementation now)

- Services table for price lists, packages, and CPT/ICD coding later.
 - Visit notes placeholder to expand to SOAP format later.
 - Payments.Method to expand to gateways in Level-2.
 - Appointments to gain reminders and wait-list in Level-2.
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